

Notes: Diamond & Van Tassel (JF, 2026)

Risk-Free Rates and Convenience Yields around the World

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1 Big picture

- **Question.** How large are convenience yields on safe assets around the world, and is the dollar special because U.S. safe assets have unusually high convenience yields or because the dollar funds international arbitrage?
- **Measurement innovation.** The paper infers risk-free rates from index option prices in 10 G11 currencies. These option-implied “box rates” come from put-call parity and box trades.
- **Main facts.** Convenience yields increase with nominal rates; U.S. Treasury convenience yields are not unusually large; box CIP deviations are similar across countries and spike in crises.
- **Takeaway.** The dollar’s special role reflects dollar-funded international arbitrage frictions more than a uniquely high dollar safe-asset convenience yield.

2 Data and measurement

The paper uses intraday index option quotes from 2004 to 2020 to infer box rates. A safe asset’s convenience yield is the box rate minus the same-currency safe asset rate. A box CIP deviation compares the U.S. box rate to a synthetic dollar box rate constructed from a foreign box rate and an FX forward. Actual box trades validate the quote-implied estimates in several markets.

3 Models and empirical specifications

3.1 Put-call parity and box rates

$$p_{i,t,T} - c_{i,t,T} = \alpha_{t,T} + \beta_{t,T}K_i + \varepsilon_{i,t,T}, \quad \beta_{t,T} = e^{-r_{t,T}T}, \quad r_{t,T} = -\frac{1}{T} \log \beta_{OLS}.$$

Detailed interpretation: The slope of the put-minus-call regression is the discount factor.

3.2 Convenience yield

$$CY_{t,T}^{safe} = r_{t,T}^{box} - r_{t,T}^{safe}.$$

3.3 CIP decomposition

$$GovCIP_{j,t} = BoxCIP_{j,t} + [CY_{j,t}^{gov} - CY_{US,t}^{gov}].$$

Detailed interpretation: Traditional government-bond CIP deviations combine international arbitrage frictions and safe-asset convenience-yield differences.

4 Identification and empirical design

The key identification assumption is that option-implied box rates are risk-free rates that do not provide safe-asset convenience. Within-currency spreads then identify safe-asset convenience yields. Cross-currency box CIP deviations identify international arbitrage balance-sheet costs because box rates do not embed safe-asset convenience yields. Trade data and tight put-call parity fits validate the box-rate measurement.

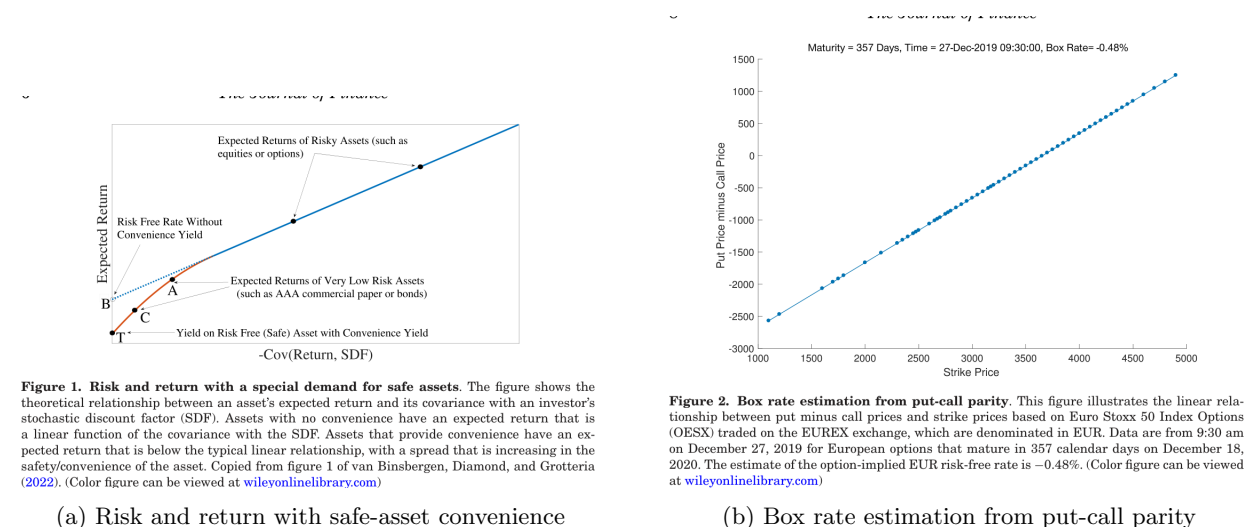
5 Tables and figures

5.1 Figure 1: Risk and return with safe-asset convenience and Figure 2: Box rate estimation from put-call parity

Read: Figure 1 motivates why a nonconvenient risk-free benchmark is needed. Figure 2 shows that put-call parity estimates a box rate from a tight linear relationship.

Detailed interpretation: The two figures justify measuring risk-free rates from options rather than from other safe assets.

Economic message: The key innovation is to estimate a convenience-yield-free benchmark.



5.2 Table I: Summary Statistics: Box Rate Sample and Precision

Read: Table I reports country coverage, sample periods, and standard errors for box-rate estimates.

Detailed interpretation: It is the data-quality table: liquid markets such as the United States and Europe have the most precise estimates.

5.3 Table II: Summary Statistics for Government Bond and Bill Convenience Yields

Read: U.S. government-bond convenience yields are positive but not the largest; bills often have high convenience yields.

Detailed interpretation: Convenience yields are international, not uniquely American.

5.4 Figure 3 and Table III

Read: Convenience yields increase with nominal government rates; the panel slope is about 12 bps per one-percentage-point higher government rate.

Table I
Summary Statistics: Box Rate Sample and Precision

Panel A presents summary statistics on the quantity of data available by currency. Panel B presents summary statistics on the standard errors for our box rate estimates. Standard errors are reported as a number of basis points. For each minute-level regression, we construct a standard error for the implied interest rate using OLS standard errors for equation (2) and the delta method.

Panel A: Sample Sizes				
Country	Stock Index	Days of Data	Start	End
United States	S & P 500	4,781	1/2/2004	7/1/2020
Europe	STOXX	4,587	1/2/2004	7/27/2020
Switzerland	Swiss Market Index	4,493	1/2/2004	7/27/2020
United Kingdom	FTSE 100	4,144	1/2/2004	7/27/2020
Sweden	OMX Stockholm 30	4,184	1/3/2005	6/30/2020
Norway	OBX 25	4,464	1/3/2005	7/27/2020
Japan	Nikkei 225	4,098	1/6/2005	7/27/2020
Australia	ASX 200	4,578	1/2/2004	7/27/2020
Canada	TSX 60	3,416	1/1/2010	3/6/2020
Denmark	OMX Copenhagen 25	2,339	1/27/2012	6/30/2020

Panel B: Standard Errors				
Country	Median Std. Err.	Mean Std. Err.	Median Std. Err.	Mean Std. Err.
United States	1.23	2.10	0.54	0.68
Europe	1.74	4.14	1.01	1.46
Switzerland	4.11	7.80	2.31	2.91
United Kingdom	7.80	16.36	3.96	4.65
Sweden	11.49	21.14	4.02	5.04
Norway	24.26	33.17	5.73	6.26
Japan	27.02	35.59		
Australia	30.48	37.71	3.59	4.24
Canada	2.50	15.70	0.41	0.71
Denmark	33.46	50.63	4.68	5.48

Maturity	>60 Days	>60 Days	>365 Days	>365 Days
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use the full panel of countries other than Japan to demonstrate robustness. (Japan is dropped because it has the fewest long-maturity option observations, resulting in noisier rate estimates.) To minimize the impact of outliers, we only use regressions with an R^2 of at least 0.99999. We discuss robustness to this

Table II
Summary Statistics for Government Bond and Bill Convenience Yields

This table presents average convenience yields relative to box rates, where 1 indicates a 1% yield. Newey-West standard errors based on a 100-day lag in parentheses. The sample averages are from January 2005 to July 2020.

Panel A: Bonds					
Country	Three Months	Six Months	One Year	Two Years	Three Years
United States	0.34 (0.03)	0.36 (0.03)	0.36 (0.03)	0.35 (0.03)	
United Kingdom	0.22 (0.08)	0.30 (0.05)	0.35 (0.03)	0.29 (0.03)	
Euro	0.29 (0.04)	0.29 (0.03)	0.28 (0.03)	0.24 (0.02)	0.21 (0.02)
Switzerland		0.00 (0.03)	0.14 (0.03)	0.19 (0.03)	
Canada	0.47 (0.09)	0.37 (0.04)	0.36 (0.02)	0.29 (0.02)	
Australia	0.61 (0.10)	0.63 (0.09)	0.60 (0.05)		

Panel B: Bills			
Country	Three Months	Six Months	One Year
United States (Bill)	0.51 (0.04)	0.46 (0.03)	0.38 (0.03)
United Kingdom (Bill)	0.23 (0.05)	0.15 (0.03)	
Euro (Bill)		0.35 (0.01)	0.30 (0.04)
Japan (Bill)	0.11 (0.09)		
Norway (Bill)	0.50 (0.04)	0.50 (0.06)	
Sweden (Bill)	0.37 (0.05)	0.48 (0.06)	
Switzerland (Bill)	0.27 (0.05)		
Canada (Bill)	0.50 (0.09)	0.38 (0.05)	0.36 (0.02)
Denmark (Bill)	0.17 (0.03)	0.15 (0.06)	

Detailed interpretation: The U.S. lies close to the fitted line rather than above it.

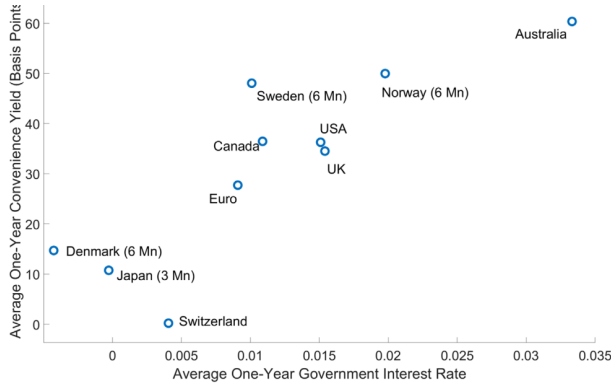


Figure 3. Cross section of average convenience yields versus average government bond rates. This figure plots the average convenience yield against the average government bond rate for each country. The convenience yield is the one-year box rate minus the one-year government bond rate in all countries except Norway, Sweden, and Denmark, where we use six-month rates, and Japan, where we use three-month rates, for a larger sample of observations to estimate average convenience yields. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Cross section of convenience yields

Table III
Convenience Yield Panel Regressions

This table presents the results of monthly panel regressions of the convenience yield onto the government bond interest rate. The sample is January 2005 to July 2020, with data from all countries except Japan. Moving blocks bootstrap standard errors with a 12-month block length are in parentheses.

	(1)	(2)	(3)	(4)
Government Rate	12.05 (1.86)	7.23 (2.14)	5.01 (2.00)	12.23 (2.41)
Local Vix				0.82 (0.24)
Observations	1,235	1,235	1,359	973
R^2	0.4236	0.1249	0.3663	0.2251
Within R^2	0.1831	0.1249	0.0723	0.2251
Fixed Effects	Month	None	Currency	None

(b) Convenience yield panel regressions

5.5 Tables IV–VI

Read: Tables IV–VI show that money-like assets have convenience yields that rise with nominal rates, that nominal rates dominate alternative predictors, and that actual box trades validate quote-implied box rates.

Detailed interpretation: The more money-like an asset is, the more its convenience yield rises with nominal rates.

Table IV
Panel Regressions of Interest Rate Spreads onto Government Bond Rates

This table reports the OLS slope coefficient and explanatory power of a panel regression of convenience yields on government bond rates for each country using different rates to estimate the convenience yield, with monthly fixed effects. For the box rate, we use a one-year maturity in all countries except Norway, Sweden, and Denmark where we use six-month rates and Japan where we use three-month rates, for a larger sample size to estimate the average convenience yield. Our corporate yield spreads are from Liao (2020) and aggregate across maturities for AUD, CAD, CHF, EUR, GBP, and JPY. We take the box rate, subtract Liao's residualized credit spread (a spread of corporate bonds over Treasuries), and add a government bond yield to get the convenience yield of corporate bonds relative to the box rate. For LIBOR and OIS, we use a one-year spread for USD, GBP, EUR, CHF, a six-month spread for SEK, DKK, CAD, AUD, and a three-month spread for JPY. For LIBOR, we use a six-month spread for NOK, and for OIS, we exclude NOK because OIS swap rates are unavailable. Savings deposits spreads are computed relative to three-month box rates. Standard errors are based on a moving blocks bootstrap with a one-year block size following Gonçalves (2011).

Dependent Variable: Box Rate Minus Listed Rate				
Rate	Slope (bps)	SE(Slope)	WithinR ²	Number of Countries
Corporate	3.06**	1.45	0.01	4
LIBOR	9.79***	1.99	0.11	10
OIS	10.24***	3.03	0.20	9
Government	15.44***	1.91	0.16	10
Government (no Japan)	12.05***	1.86	0.18	9
Deposit	71.90***	7.10	0.59	10

(a) Table IV

Table V
Cross-Country Regressions of Average Treasury Convenience Yields onto Different Explanatory Factors

Each row reports the slope from a cross-sectional OLS regression with an intercept of the average box convenience yield on the average explanatory factor across countries. The last row reports the number of countries in the regression (N).

	Slope (bps)	R ²	# Countries
Panel A: Nominal Interest Rates			
Government Bond 1y	15.63	0.74	10
Government Bond 2y	19.04	0.77	10
Central Bank Policy Rate	16.50	0.72	10
LIBOR 3m	16.63	0.77	10
Panel B: Credit Risk and Bond Supply			
CDS 5y	-0.31	0.15	10
CDS 1y	-1.49	0.09	10
Debt-to-GDP Ratio	-0.10	0.09	10
Panel C: Frictions and Volatility			
Normalized Option B-spread	-3.29	0.03	10
LIBOR minus OIS spread	-0.49	0.16	9
Stock market realized volatility	1.27	0.04	10
VIX	-1.22	0.03	10

(b) Table V

Table VI
Summary Statistics: Average Convenience Yields for Trade Data

Each convenience yield is the spread between the interest rate on a box trade and a maturity-matched government bond yield on the same day. The "matched quotes" column uses the quote-data-implied box rate for that day instead of the box trade. We report the mean of each country's convenience yield, with its standard deviation (not a standard error) and number of observations in parentheses.

Country	Mean, Trades	Mean, Matched Quotes	Start	End
United States	0.283 (0.16, 2845)	0.282 (0.15, 2840)	5/18/2010	7/1/2020
Europe	0.31 (0.30, 3559)	0.32 (0.26, 3452)	12/14/2009	7/21/2020
United Kingdom	0.31 (0.35, 87)	0.33 (0.36, 86)	1/8/2015	7/22/2020
United Kingdom (vs. OIS)	0.23 (0.32, 290)	0.21 (0.30, 287)	1/8/2015	7/22/2020
Switzerland	-0.098 (0.16, 155)	-0.060 (0.21, 135)	1/1/2010	3/30/2020
Australia	1.20 (0.66, 933)	0.60 (2.95, 877)	5/18/2010	7/17/2020

one. See Section III of the Internet Appendix for details on measurement and more results using the trade data.

(c) Table VI

5.6 Figure 4, Table VII, Figure 5, and Table VIII

Read: These visuals compare box CIP deviations to traditional CIP deviations. Box CIP deviations are similar across countries, while government-bond CIP deviations vary with nominal rates.

Detailed interpretation: Traditional CIP deviations embed convenience-yield differences; box CIP deviations isolate international arbitrage costs.

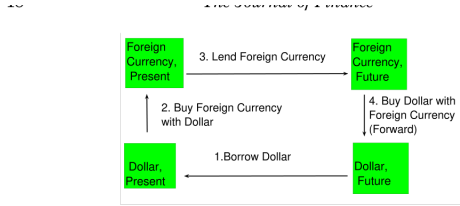


Figure 4. Trading strategy to exploit CIP deviations. (Color figure can be viewed at [wiley-onlinelibrary.com](https://onlinelibrary.com))

(a) Figure 4

Table VII
Summary Statistics for Average Box Rate CIP Deviations

A deviation of 1 represents a 1% yield spread, where the sign convention is that a negative spread reflects that dollar box rates are lower than a rate constructed with foreign box rates and foreign exchange transactions. Newey-West standard errors with a 100-day lag in parentheses. Sample is from Jan 2005 to July 2020.

	Three Months	Six Months	One Year	Two Years
Panel A: Box Rates				
Australia	-0.08 (0.11)	-0.10 (0.10)	-0.09 (0.08)	
Canada	-0.19 (0.10)	-0.12 (0.06)	-0.11 (0.02)	
Switzerland	-0.08 (0.03)	-0.11 (0.03)	-0.14 (0.04)	-0.20 (0.04)
Euro	-0.04 (0.03)	-0.05 (0.03)	-0.08 (0.03)	-0.12 (0.03)
United Kingdom	0.10 (0.05)	-0.01 (0.03)	-0.04 (0.03)	-0.06 (0.08)
Japan	-0.14 (0.14)			
Norway	-0.12 (0.05)	-0.11 (0.06)		
Sweden	-0.13 (0.05)	-0.23 (0.06)		
Denmark	-0.08 (0.04)	-0.13 (0.04)		
Panel B: Gov Bonds				
Australia	0.07 (0.04)	0.04 (0.04)	0.07 (0.04)	
Canada	0.02 (0.02)	-0.03 (0.01)	-0.06 (0.01)	0.01 (0.04)
Switzerland			-0.47 (0.04)	-0.37 (0.05)
Euro	-0.12 (0.02)	-0.14 (0.02)	-0.18 (0.03)	-0.22 (0.05)
United Kingdom	-0.19 (0.05)	-0.13 (0.03)	-0.06 (0.02)	-0.01 (0.005)
Japan	-0.32 (0.03)	-0.36 (0.03)	-0.44 (0.03)	
Norway	-0.02 (0.03)	-0.06 (0.04)		
Sweden	-0.10 (0.05)	-0.11 (0.04)		
Denmark	-0.20 (0.03)	-0.28 (0.04)	-0.44 (0.03)	

are larger than for safe assets in other currencies, then the second term should be negative.

B. CIP Deviation Results

(b) Table VII

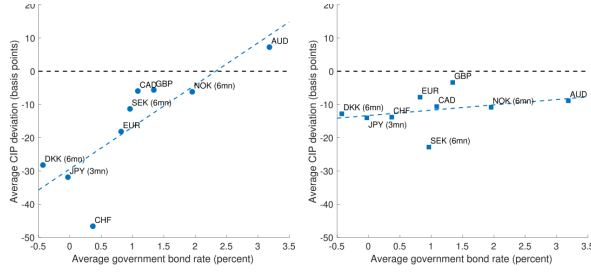


Figure 5. Cross section of average CIP deviations for box rates and government bond rates. This figure plots the average CIP deviation for government bond rates and box rates against the average government bond rate for each country. The CIP deviation is computed for a one-year maturity in all countries except Norway, Sweden, and Denmark, where we use six-month rates, and Japan, where we use three-month rates, for a larger sample size of box rates from which we estimate the average CIP deviations. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 5

Table VIII
Cross-Country/Panel Regressions of CIP Deviations onto Government Bond Rates

This table presents regressions of CIP deviations computed using different interest rates on government bond rates. For the box rate, we use a one-year maturity in all countries except Norway, Sweden, and Denmark where we use six-month rates, and Japan, where we use three-month rates for a larger sample size. Our panel regression excludes Japan and includes monthly fixed effects. For the corporate rate, we use the corporate basis from Liao (2020), which is available for AUD, CAD, CHF, EUR, GBP, and JPY. For LIBOR and OIS, we use a one-year spread for USD, GBP, EUR, and CHF, a six-month spread for SEK, DKK, CAD, and AUD, and a three-month spread for JPY. For LIBOR, we use a six-month spread for NOK, and for OIS, we exclude NOK because OIS swap rates are unavailable. The last column reports the number of countries in the regression (N). One-year moving block bootstrap standard errors are in parentheses for the Slope (panel) column.

CIP deviation	Slope (Cross Section)	Slope (Panel)	R^2 (Cross Section)	Within R^2 (Panel)	N
Box	1.60	0.89 (1.38)	0.10	0.0008	9
Corporate	-1.97	-1.96 (1.29)	0.14	0.050	6
LIBOR	12.21	6.18 (1.74)	0.64	0.109	9
OIS	14.60	11.07 (3.23)	0.82	0.112	8
Government	12.65	10.56 (1.92)	0.67	0.242	9
Deposit	49.39	56.87 (4.28)	0.51	0.521	9

(b) Table VIII

5.7 Figures 6–8 and Tables IX–XI

Read: Figures 6–8 and Tables IX–XI show crisis behavior and exchange-rate relationships. Convenience yields rise in domestic crises; box CIP deviations rise in all crises; broad dollar appreciation predicts larger box CIP deviations.

Detailed interpretation: Crisis CIP spikes are primarily international-arbitrage-friction events, not evidence that U.S. safe assets become uniquely convenient.

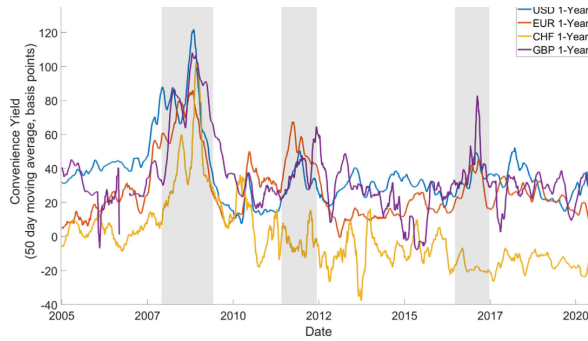


Figure 6. Time series of convenience yields for USD, EUR, CHF, and GBP. This figure plots the time series of box convenience yield estimates for a one-year maturity for the United States, Europe, Switzerland, and the United Kingdom as a 50-day moving average in basis points. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 6

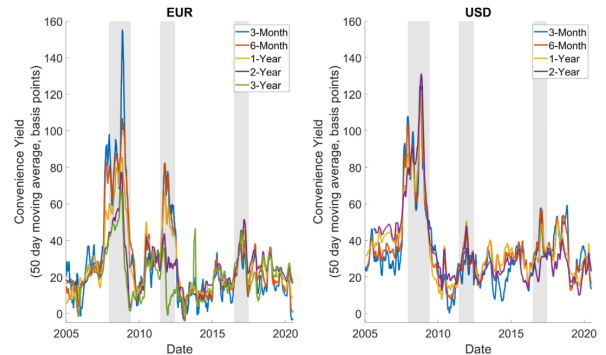


Figure 7. Term structure of convenience yields for EUR and USD. This figure plots the

(b) Figure 7

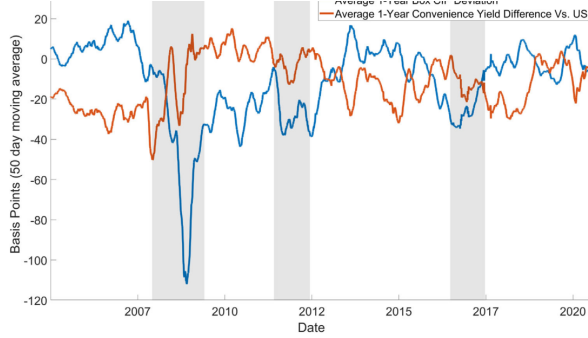


Figure 8. Average CIP deviation versus average convenience yield difference from the United States for all currencies except JPY. This figure plots a time series of CIP deviations and a time series of convenience yield differences. The CIP deviation and convenience yield difference are averaged across available countries each day and then reported as a 50-day moving average. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 8

Table X
Monthly Exchange Rate Forecast Regressions, Following Engel and Wu (2023)

This table presents monthly regressions from January 2004 to July 2020, following the specification in equation (12). Each regression attempts to predict either next month's log exchange rate change $\Delta s_{j,t+1}$ or the contemporaneous exchange rate change $\Delta s_{j,t}$. Regressions are run both on USD/GBP/EUR/CHF only and then for all currencies except JPY. Panel B presents the same regressions as in Panel A, except the predictor variables $\Delta \text{BoxCIP}_{j,t}$, $\Delta \text{CYdiff}_{j,t-1}$ for each country are replaced by their cross-sectional average across all non-JPY countries in the same month.

Panel A:		Panel Regression			
Predicted Variable	$\Delta s_{j,t+1}$	$\Delta s_{j,t}$	$\Delta s_{j,t+1}$	$\Delta s_{j,t}$	
$\Delta \text{BoxCIP}_{j,t}$	-5.13 (2.00)	-4.27 (2.03)	-3.81 (1.45)	-5.66 (1.68)	
$\text{BoxCIP}_{j,t-1}$	-0.862 (0.814)	-1.79 (0.803)	-0.543 (0.718)	-1.70 (0.734)	
$\Delta \text{CYdiff}_{j,t}$	-3.88 (1.80)	-4.17 (2.20)	-3.14 (1.29)	-5.21 (1.61)	
$\Delta \text{CYdiff}_{j,t-1}$	-0.836 (0.937)	-1.86 (1.12)	-0.543 (0.548)	-1.43 (0.593)	
R-squared:	0.0531	0.0626	0.0385	0.0836	
Currencies:	USD/GBP/EUR/CHF	USD/GBP/EUR/CHF	All but JPY	All but JPY	

Panel B:		Averaged Predictors			
Predicted Variable	$\Delta s_{j,t+1}$	$\Delta s_{j,t}$	$\Delta s_{j,t+1}$	$\Delta s_{j,t}$	
$\Delta \text{BoxCIP}_{j,t}$	-6.77 (2.36)	-5.93 (2.41)	-6.66 (2.44)	-9.28 (2.60)	
$\text{BoxCIP}_{j,t-1}$	-1.28 (1.17)	-2.64 (.986)	-1.00 (1.54)	-2.60 (1.46)	
$\Delta \text{CYdiff}_{j,t}$	-5.08 (2.57)	-6.24 (2.93)	-5.19 (2.28)	-8.31 (2.37)	
$\Delta \text{CYdiff}_{j,t-1}$	-0.929 (1.64)	-2.91 (1.81)	-0.647 (1.41)	-2.56 (1.27)	
R ² :	0.0643	0.0825	0.0555	0.123	
Currencies:	USD/GBP/EUR/CHF	USD/GBP/EUR/CHF	All but JPY	All but JPY	

lated with the broad dollar exchange rate. Under the interpretation of Avdjiev et al. (2019) that the broad dollar is a good barometer for global financial frictions, this provides additional evidence that box CIP deviations spike during financial crises but that convenience yield differences do not.

We use changes in the U.S. trade-weighted broad dollar index ΔDollar_t and changes in country j 's bilateral dollar exchange rate ΔBER_j to explain contemporaneous changes $\Delta x_{j,t}$ in the variable $x_{j,t}$,

$$\Delta x_{j,t} = \alpha_j + \beta \Delta \text{Dollar}_t + \gamma \Delta \text{BER}_j + \epsilon_{j,t}. \quad (13)$$

(a) Table X

Table IX
Regressions of Box CIP Deviations and Convenience Yield Differences on Government CIP Deviations

The first column presents time-series regression results of the average one-year box CIP deviation for the United Kingdom, Europe, and Switzerland on the average one-year government CIP deviation for these same three countries. The second column presents time-series regression results of the difference in the average one-year convenience yield of these three countries on the average of their one-year government yield CIP deviations. The third and fourth columns repeat these regressions using all countries except Japan. Observations for a country are included only when all of its box CIP, government CIP, and convenience yield difference observations are not missing. Observations are at a daily frequency, with Newey-West standard errors presented using a 100-day lag. The sample is from Jan 2005 to July 2020.

Dependent Var.	Box CIP	Conv. Yield Difference	Box CIP	Conv. Yield Difference
Intercept	0.0011 (0.0003)	-0.0011 (0.0003)	0.0008 (0.0003)	-0.0008 (0.0003)
Slope	0.8478 (0.0747)	0.1522 (0.0747)	0.8124 (0.0972)	0.1876 (0.0972)
R ²	0.503	0.032	0.421	0.037
Currencies	EUR/CHF/GBP	EUR/CHF/GBP	All but JPY	All but JPY

(b) Table IX

Table XI
Regression of Box CIP Deviation (Panel A, in Basis Points) and Convenience Yield Differences (Panel B, in Basis Points) on Bilateral Exchange Rate and Broad Dollar Exchange Rate

Monthly Regressions Are from January 2004 to July 2020.

Panel A: CIP Deviations			Panel B: Convenience Yield Differences		
USD/GBP/EUR/CHF			USD/GBP/EUR/CHF		
ΔBER_j	0.229 (0.35)	-0.47 (0.23)	ΔBER_j	-0.27 (0.54)	-0.22 (0.22)
ΔDollar_t	-1.72 (0.56)		ΔDollar_t	-0.15 (0.34)	
R ²	0.027	0.007	R ²	0.003	0.002
All But JPY			All But JPY		
ΔBER_j	-0.12 (0.35)	-0.84 (0.23)	ΔBER_j	-0.62 (0.54)	-0.26 (0.22)
ΔDollar_t	-1.78 (0.56)		ΔDollar_t	0.78 (0.34)	
R ²	0.015	0.008	R ²	0.002	0.0009

associated with a 1.72 bp (or 1.78 bp, depending on the sample of countries) increase in the size of the one-year box CIP deviation. This estimate is between the 2.6 and 1.1 bps, respectively, reported at a three-month and five-year maturity in Avdjiev et al. (2019). Relative to existing work, our contribution is to show that the relationship between CIP deviations and the dollar is due to international arbitrage frictions measured by box CIP deviations and not to differences between countries' convenience yields.

(b) Table XI

6 Short summary

- The paper measures international risk-free rates using option-implied **box rates**. A box trade creates a risk-free payoff using options, and the implied discount rate is inferred from put-call parity.
- The key measurement problem is that many assets commonly called “risk-free” also provide convenience services. Government bonds, bills, deposits, repo, and other safe assets may be liquid, pledgeable, regulatory-friendly, or money-like. Their yields can therefore be below the pure risk-free rate.
- The paper uses box rates because they are meant to be risk-free rates without the same safe-asset convenience services. This allows the authors to separate the pure risk-free rate from the safe-asset convenience

yield.

- The basic convenience-yield measure is

$$CY_{t,T}^{safe} = r_{t,T}^{box} - r_{t,T}^{safe}.$$

A positive convenience yield means investors accept a lower yield on the safe asset than on the option-implied box rate because the safe asset provides extra nonpecuniary services.

- The paper studies convenience yields in 10 G11 currencies using intraday index option data from 2004 to 2020. This gives an international comparison of safe-asset convenience yields rather than a U.S.-only measurement.
- A central fact is that convenience yields exist across many safe sovereign debt markets. U.S. Treasuries are convenient, but other safe sovereign assets are also convenient.
- Convenience yields increase with nominal interest rates. The paper finds that a one-percentage-point higher nominal government rate is associated with roughly a 12–15 basis point higher government-debt convenience yield.
- The United States is not uniquely high in government-bond convenience yields. The U.S. Treasury convenience yield is positive, but it is not dramatically larger than the convenience yields of other safe sovereign issuers.
- This result changes the interpretation of dollar specialness. The dollar remains special, but the specialness is not mainly that U.S. Treasuries provide uniquely high convenience yields.
- The paper also studies covered interest parity deviations. Traditional government-bond CIP deviations mix two different forces:

$$GovCIP_{j,t} = BoxCIP_{j,t} + \left(CY_{j,t}^{gov} - CY_{US,t}^{gov} \right).$$

The first term is the international arbitrage wedge, and the second term is the difference between foreign and U.S. safe-asset convenience yields.

- Box CIP deviations are cleaner measures of cross-currency arbitrage frictions because box rates do not embed government-bond convenience yields.
- The paper finds that average box CIP deviations are similar across countries. This supports the idea that box CIP deviations measure the common balance-sheet cost faced by international arbitrageurs.
- During crises, box CIP deviations widen broadly across countries. This suggests that global intermediary balance-sheet constraints and dollar funding frictions become more severe in stress periods.
- Convenience yields also rise during crises, but they tend to rise most in the country where the crisis is centered. U.S. convenience yields are not always the ones that rise the most.
- The paper therefore separates two crisis channels: domestic safe-asset demand raises convenience yields, while global arbitrage stress raises box CIP deviations.
- Exchange-rate evidence supports the arbitrage-friction interpretation. Broad dollar appreciation is associated with larger box CIP deviations, suggesting that dollar strength is linked to tighter global dollar funding and arbitrage constraints.
- The paper’s main takeaway is that safe-asset convenience yields are international and rate-sensitive, while the dollar’s special role is tied more to global funding and international arbitrage than to uniquely high Treasury convenience yields.

7 Conclusion

7.1 Core conclusion

Diamond and Van Tassel show that safe-asset convenience yields are widespread across countries. U.S. Treasuries are convenient assets, but their convenience yield is not uniquely large once measured against option-implied box rates.

The paper’s central conclusion is that the dollar’s special role in global finance is better understood through global funding and cross-currency arbitrage frictions than through an unusually large convenience yield on U.S. government debt.

7.2 Measurement contribution

The paper’s key measurement contribution is the construction of international risk-free rates from option-implied **box rates**. A box rate is inferred from put-call parity and represents the discount rate on a risk-free payoff created using options.

This matters because many conventional risk-free benchmarks are not pure risk-free rates. Government bonds, Treasury bills, repo rates, deposits, and other safe assets can all embed convenience yields.

The paper’s measurement framework is:

$$CY_{t,T}^{safe} = r_{t,T}^{box} - r_{t,T}^{safe}.$$

This separates:

- the pure risk-free rate, measured by the box rate;
- the observed safe-asset yield;
- and the convenience yield embedded in the safe asset.

7.3 Economic intuition

Safe assets provide services beyond their cash flows. Investors may value them because they are liquid, safe, pledgeable, useful as collateral, or useful for regulatory and institutional purposes.

Because of these services, investors accept lower yields on safe assets. The gap between the box rate and the safe-asset rate measures how much yield investors give up to obtain those services.

The paper shows that this logic applies internationally. Safe-asset convenience is not only a U.S. Treasury phenomenon. Other safe sovereign debt markets also provide convenience.

7.4 Nominal rates and convenience yields

A major result is that convenience yields rise with nominal interest rates. When nominal rates are higher, the opportunity cost of holding money-like or safe assets is higher, so the convenience yield becomes more visible.

This is especially important for more money-like assets, such as bills and highly liquid short-term instruments. The paper’s evidence suggests that nominal rates are a key driver of cross-country differences in convenience yields.

7.5 Dollar specialness

The paper does not deny that the dollar is special. Instead, it clarifies what kind of specialness the dollar has.

The evidence does not support the view that U.S. safe assets have uniquely high convenience yields. Instead, the dollar’s special role comes from its central position in international funding and arbitrage markets.

In other words, dollar specialness appears more clearly in global arbitrage wedges, such as box CIP deviations, than in the domestic convenience yield of U.S. Treasuries.

7.6 CIP interpretation

Traditional government-bond CIP deviations are hard to interpret because government bonds have convenience yields. A government-bond CIP deviation combines international arbitrage frictions with cross-country differences in safe-asset convenience yields.

The paper’s key decomposition is:

$$GovCIP_{j,t} = BoxCIP_{j,t} + \left(CY_{j,t}^{gov} - CY_{US,t}^{gov} \right).$$

This equation shows that government-bond CIP deviations are not pure arbitrage wedges. They also reflect whether the foreign safe asset has a different convenience yield from the U.S. safe asset.

Box CIP deviations are cleaner because box rates do not embed government-bond convenience services. Therefore, box CIP deviations better isolate international arbitrage and balance-sheet frictions.

7.7 Crisis interpretation

During crises, two things can happen at the same time. First, investors may demand more local safe assets, raising domestic convenience yields. Second, international arbitrageurs may become balance-sheet constrained, raising box CIP deviations.

The paper shows that convenience yields rise most in the country where the crisis is centered. For example, U.S. convenience yields rise strongly in the global financial crisis, euro convenience yields rise during the euro crisis, and U.K. convenience yields rise around Brexit stress.

By contrast, box CIP deviations rise broadly across currencies during crises. This suggests that crisis-time CIP spikes are mainly about global arbitrage frictions, not simply about U.S. safe assets becoming uniquely convenient.

7.8 Exchange-rate interpretation

The exchange-rate evidence supports the global-arbitrage view. When the broad dollar strengthens, box CIP deviations widen. This implies that dollar appreciation is associated with tighter global dollar funding and larger arbitrage wedges.

Convenience-yield differences are less closely tied to broad dollar movements. This reinforces the paper’s claim that dollar specialness is mainly about funding and arbitrage, not unusually high Treasury convenience yields.

7.9 Policy implications

The paper implies that policymakers should distinguish between domestic safe-asset scarcity and international arbitrage stress.

A rise in a country’s government-bond convenience yield may reflect local demand for safe assets, collateral, or liquidity. A rise in box CIP deviations reflects stress in international arbitrage and funding markets.

These are different problems. Domestic safe-asset scarcity may be addressed by safe-asset supply or collateral policy. International arbitrage stress may require liquidity facilities, swap lines, or policies that relax intermediary balance-sheet constraints.

7.10 Limitations

The box-rate method requires liquid option markets. The estimates are most reliable for countries and maturities with deep option markets and tight put-call parity relationships.

Box rates are also option-market-implied rates, so they may reflect features of option-market clearing, margining, and trading frictions. The paper addresses this concern with actual box-trade evidence, but measurement noise may still be larger in less liquid markets.

The sample focuses on G11 currencies and developed safe-asset markets. The conclusions may not directly apply to emerging markets or countries with less liquid derivatives and sovereign bond markets.

7.11 Takeaway

The paper's main lesson is that researchers should separate three objects:

pure risk-free rate, safe-asset convenience yield, international arbitrage wedge.

Box rates make this separation possible. Once the decomposition is made, U.S. safe assets no longer look uniquely convenient. The dollar's special role instead appears in global funding markets and cross-currency arbitrage frictions.